

Shubham Singh

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EDUCATION

Bachelor of Technology (B.Tech) - Computer Science and Engineering

2023 - Present

Noida Institute of Engineering and Technology, Greater Noida, UP

CGPA: 8.05

PROJECTS

OrbitWatcher - Satellite Collision Risk Intelligence Platform *[Live Demo]* *[GitHub]* Mar 2026 - Aug 2026

Python, FastAPI, SGP4, SciPy KDTree, SQLite, Three.js, Docker

- Engineered an end-to-end orbital monitoring platform that processes TLE data for 10,000+ Starlink satellites.
- Optimized conjunction screening using SciPy KDTree, achieving an 11,000+ $\times$ benchmarked speedup over brute-force pairwise screening.
- Developed and deployed a containerized FastAPI application with collision risk analysis, forecasting, maneuver recommendations, and 3D visualization.

Safar Sathi - Smart India Hackathon 2025 (Team Leader) *[Live Demo]* *[GitHub]*

Sep 2025 - Oct 2025

React.js, Node.js, Express.js, MongoDB

- Developed and deployed a full-stack tourism web application using React.js, Node.js, Express.js, and MongoDB with OTP-based authentication for tourists and local guides.
- Developed trip-planning features including accommodation cost estimation and destination recommendations.
- Integrated an AI-powered chatbot to enhance user engagement and assist with travel-related queries.

EXPERIENCE

Cyfuture India Pvt. Ltd.

Jul 2026 - Sep 2026

Intern, Application Development - AI/ML, Onsite

- Gained hands-on experience in application development in an AI/ML-focused development environment, understanding project workflows, team collaboration, and software delivery practices.
- Worked on full-stack AI/ML applications involving bot and agent integration, REST API consumption, machine learning workflows, and deployment.
- Developed practical skills in Git/GitHub, cloud fundamentals, API integration, and selecting appropriate machine learning algorithms for application use cases.

Software Engineering Virtual Experience - Accenture (Forage) *[Certificate]*

Jul 2025 - Aug 2025

Intern, Remote

- Learned Agile and SDLC concepts through software engineering simulations focused on requirements.
- Analyzed requirements, identified defects, and performed debugging activities to improve software reliability.

TECHNICAL SKILLS

Languages: Python, C++

Frontend: HTML, CSS, React.js

Backend: Node.js, Express.js, FastAPI, REST APIs

Databases: MongoDB, MySQL

Cloud & ML: Cloud Fundamentals, Machine Learning

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, DBMS

Development & Deployment: Git, GitHub, Docker, Render, Vercel

Soft Skills: Adaptability, Leadership, Communication, Collaboration, Problem-solving

COMPETITIVE PROGRAMMING

- Solved 300+ Data Structures and Algorithms problems on LeetCode.
- Achieved a LeetCode Contest Rating of 1500+ through consistent participation in weekly and biweekly contests.
- Strong understanding of Arrays, Strings, Linked Lists, Recursion, Hash Table, Sliding Window and Trees.

CERTIFICATIONS

- **Design Thinking:** Learned user-centered problem-solving using design thinking.
- **Data Structures and Algorithms:** Developed problem-solving skills through practice of DSA concepts.
- **Object Oriented Programming:** Practiced OOP principles for structured program design.
- **Introduction to MongoDB:** Practiced database design and CRUD operations in MongoDB.